Max Heap Sort

Definition

Max Heap Sort is a sorting algorithm that uses a **Max Heap** data structure. In a Max Heap, the **largest element is always at the root**. The largest element is repeatedly moved to the end of the array, producing a sorted array in **ascending order**.

Algorithm

1. Build a **Max Heap** from the given array.
2. The largest element is now at the root.
3. Swap the root element with the last element.
4. Reduce the heap size by 1.
5. Apply **heapify** to restore the Max Heap property.
6. Repeat steps 3–5 until only one element remains.
7. The array is now sorted in ascending order.

Output:

```
Enter number of elements: 5
Enter elements: 7
8
8
6
2
Sorted array: 2 6 7 8 8

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

Case	Time Complexity
Best Case	**O(n log n)**
Average Case	**O(n log n)**
Worst Case	**O(n log n)**

Building the Max Heap: O(n)

Repeated heapify operations: O(n log n)

Space Complexity

O(1) auxiliary space because Max Heap Sort can sort the array **in-place**.